21/5/26

Multiple Linear Regression

$$\theta = (X^T X)^{-1} X^T Y$$

$$X = \begin{bmatrix} 1 & 5 & 11 & 125 \\ 1 & 7 & 9 & 155 \\ 1 & 9 & 15 & 175 \\ 1 & 11 & 13 & 220 \\ 1 & 13 & 18 & 285 \end{bmatrix} \quad Y = \begin{bmatrix} 48 \\ 72 \\ 101 \\ 142 \\ 176 \end{bmatrix}$$

$$X^T X = \begin{bmatrix} 5 & 45 & 66 & 910 \\ 45 & 445 & 630 & 8760 \\ 66 & 630 & 920 & 12485 \\ 910 & 8760 & 12485 & 173900 \end{bmatrix}$$

$$X^T Y = \begin{bmatrix} 539 \\ 5503 \\ 7705 \\ 107435 \end{bmatrix}$$

'Calculations have been done using scientific calculator'.

since $\det(X^T X) \neq 0$

$$\det(X^T X) \cong 2.86 \times 10^7$$

let $A = X^T X \Rightarrow A^{-1} = \frac{1}{|A|}(\text{Adj}\,A)$

$\Rightarrow$

$$A^{-1} = \begin{bmatrix} 76.9017 & 18.2148 & -3.0043 & -1.1043 \\ 18.2148 & 4.5863 & -0.7241 & -0.2744 \\ -3.0043 & -0.7241 & +0.1602 & 0.0407 \\ -1.1043 & -0.2744 & 0.0407 & 0.0167 \end{bmatrix}$$

$$\theta = (X^T X)^{-1} X^T Y$$

$$\theta = \begin{bmatrix} -99.911 \\ 1.217 \\ 8.417 \\ 0.906 \end{bmatrix}$$

classmate

Date

Page

Final Regression equation

$$y = -99.911 + 1.217x_1 + 2.417x_2 + 0.906x_3$$

where x_1, x_2 and x_3 are independent variables and y is the dependent variables.

$x_1 \rightarrow$ windspeed

$x_2 \rightarrow$ Blade angle

$x_3 \rightarrow$ Rotor speed

$y \rightarrow$ Power output

given $\rightarrow$ 9, 11, 100

$$\rightarrow y = -99.911 + 1.217(9) + 2.417(11) + 0.906(100)$$

$$= -99.911 + 10.953 + 26.587 + 90.6 \Rightarrow \boxed{y = 28.23}$$

$$\{y = 28.2290 \cong 28.23\}$$